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Arthur Cayley

851 (continued). On Linear Differential Equations.

(Continuation from p. 405; PDF page 426.)

of four factors, and assuming

$$\begin{aligned} D + \delta &= (1, p_1 | D, 1), \\ (D + \gamma)(D + \delta) &= (1, p_2, q_2 | D, 1)^2, \\ (D + \beta)(D + \gamma)(D + \delta) &= (1, p_3, q_3, r_3 | D, 1)^3, \\ (D + \alpha)(D + \beta)(D + \gamma)(D + \delta) &= (1, p_4, q_4, r_4, s_4 | D, 1)^4, \end{aligned}$$

we have the sets of equations

$$\begin{aligned} (1) \quad & p_1 = \delta. \\ (2) \quad & p_2 = \gamma + p_1, \\ & q_2 = \gamma p_1 + p'_1. \\ (3) \quad & p_3 = \beta + p_2, \\ & q_3 = \beta p_2 + p'_2 + q_2, \\ & r_3 = \beta q_2 + q'_2. \\ (4) \quad & p_4 = \alpha + p_3, \\ & q_4 = \alpha p_3 + p'_3 + q_3, \\ & r_4 = \alpha q_3 + q'_3 + r_3, \\ & s_4 = \alpha r_3 + r'_3. \end{aligned}$$

7. I stop to remark that there would have been a convenience in considering the product

$$\left(\frac{d}{dx} + \delta\right)\left(\frac{d}{dx} + \gamma\right)\left(\frac{d}{dx} + \beta\right)\left(\frac{d}{dx} + \alpha\right),$$

for the first, second, third, &c., sets of equations would then have contained α , β , γ , &c., respectively; but for better comparison with the equations of the reverse order, I have preferred not to make this change of notation.

8. The set (1) gives $\delta - p_1 = 0$.

Set (2) gives

$$p_2 = -\gamma + p_1, \quad 0 = \gamma' + \gamma^2 - p_1\gamma - p'_1 + q_2.$$

Set (3) gives

$$p_3 = -\beta + p_2, \quad 0 = \beta' + \beta^2 - p_2\beta - p'_2 + q_2 + (\beta' + \beta^2)q'_2 - p_2q_2 - r_2 \cdots$$

$$q_3 = \beta p_2 + p'_2 + q_2, \quad r_3 = \beta q_2 + q'_2.$$

Set (4) gives

$$\begin{aligned}
 p_4 &= -\alpha + p_3, \\
 q_4 &= \alpha^2 + 3\alpha\alpha' + \alpha'' - p_3\alpha - p_3' + p_2(\alpha^2 + \alpha') + 2p_2'\alpha + p_2'', \\
 0 &= \alpha''' + 4\alpha\alpha'' + 6\alpha'^2 + 3\alpha^2\alpha' + \alpha^4 \\
 &\quad - p_3(\alpha'' + 3\alpha\alpha' + \alpha^3) + p_2(3\alpha' + 3\alpha^2) - 3p_3''\alpha \\
 &\quad - p_3'(3\alpha + \alpha^2) - 3p_4''' \\
 &\quad + q_2(\alpha^2 + \alpha') + 2q_2'\alpha + q_2'' \\
 &\quad - r_3'\alpha - r_4' \\
 &\quad + s_4,
 \end{aligned}$$

which differ in form from the equations belonging to the reverse order in containing the derived functions $p_2', p_2'', p_3', p_3'', q_2', q_2'', \dots$ of the coefficients.

9. Taking p_4, q_4, r_4, s_4 as known, we have α determined by a differential equation (not linear) of the third order; and α being known, we know p_3, q_3, r_3 . We then have β determined by a differential equation (not linear) of the second order; and β being known, we know p_2, q_2 . We then have γ determined by a differential equation (not linear) of the first order; and γ being known, we know p_1 , that is, δ .

852. Note sur le Mémoire de M. Picard “Sur les Intégrales de Différentielles Totales Algébriques de Première Espèce.”

[From the *Bulletin des Sciences Mathématiques*, 2^{me} Sér., t. x. (1886), pp. 75–78.]

ON peut présenter l'analyse sur laquelle est fondé le Mémoire sous une forme plus symétrique en introduisant dès le commencement les fonctions homogènes.

Soit $f = (*) (x, y, z, t)^m$ une fonction du degré m des variables x, y, z, t , lesquelles seront toujours liées par l'équation $f = 0$: écrivons aussi

$$\frac{df}{dx}, \frac{df}{dy}, \frac{df}{dz}, \frac{df}{dt} = X, Y, Z, T,$$

de manière que X, Y, Z, T sont des fonctions du degré $m - 1$: et soient A, B, C, D des fonctions chacune du degré $m - 3$ et Q une fonction du degré $m - 4$, telles que

$$AX + BY + CZ + DT = Qf \quad \text{identiquement;}$$

donc, en supposant $f = 0$, on aura

$$AX + BY + CZ + DT = 0.$$

On vérifie sans peine que l'expression

$$d\Omega = \frac{1}{\lambda X + \mu Y + \nu Z + \rho T} \begin{vmatrix} \lambda, & \mu, & \nu, & \rho \\ A, & B, & C, & D \\ x, & y, & z, & t \\ dx, & dy, & dz, & dt \end{vmatrix}$$

est indépendante des valeurs de λ, μ, ν, ρ , et ainsi égale à chacune des quatre expressions $d\Omega_x, d\Omega_y, d\Omega_z, d\Omega_t$,

$$\begin{aligned} \frac{1}{X} \begin{vmatrix} B, & C, & D \\ y, & z, & t \\ dy, & dz, & dt \end{vmatrix}, & \quad -\frac{1}{Y} \begin{vmatrix} A, & C, & D \\ x, & z, & t \\ dx, & dz, & dt \end{vmatrix}, \\ \frac{1}{Z} \begin{vmatrix} A, & B, & D \\ x, & y, & t \\ dx, & dy, & dt \end{vmatrix}, & \quad -\frac{1}{T} \begin{vmatrix} A, & B, & C \\ x, & y, & z \\ dx, & dy, & dz \end{vmatrix}, \end{aligned}$$

respectivement.

Cela étant,

$$d\Omega_t = -\frac{1}{T} \begin{vmatrix} A, & B, & C \\ x, & y, & z \\ dx, & dy, & dz \end{vmatrix} = \text{une différentielle totale;}$$

en écrivant un moment $x, y, z = x't, y't, z't$, et en dénotant par f' la fonction $(*)(x', y', z', 1)^m$, et de même par $X', Y', Z', T', A', B', C'$ les valeurs correspondantes de X, Y, Z, T, A, B, C , les variables x', y', z' seront liées par l'équation $f' = 0$, ce qui donne

$$X'dx' + Y'dy' + Z'dz' = 0;$$

et l'on voit sans peine que l'expression de $d\Omega_t$ se réduit à

$$-\frac{1}{T'} \begin{vmatrix} A', & B', & C' \\ x', & y', & z' \\ dx', & dy', & dz' \end{vmatrix},$$

fonction de la forme

$$F'dx' + G'dy' + H'dz',$$

qui ne contient que les variables x', y', z' . Donc, en omettant les accents, il est permis de prendre $t = \text{const.}$, ce qui donne

$$X dx + Y dy + Z dz = 0;$$

et avec cette relation entre dx, dy, dz , la forme ci-dessus pour $d\Omega_t = F dx + G dy + H dz$ soit une différentielle totale: cela donne la condition

$$X\left(\frac{dG}{dx} - \frac{dH}{dy}\right) + Y\left(\frac{dH}{dx} - \frac{dF}{dz}\right) + Z\left(\frac{dF}{dy} - \frac{dG}{dx}\right) = 0,$$

ou enfin

$$\begin{aligned} & X\left(\frac{d}{dx} \frac{Cx - Az}{T} - \frac{d}{dy} \frac{Ay - Bx}{T}\right) \\ & + Y\left(\frac{d}{dx} \frac{Ay - Bx}{T} - \frac{d}{dz} \frac{Bz - Cy}{T}\right) \\ & + Z\left(\frac{d}{dy} \frac{Bz - Cy}{T} - \frac{d}{dx} \frac{Cx - Az}{T}\right) = 0. \end{aligned}$$

On a d'abord un terme $\mathfrak{A} = \frac{1}{T}$, où

$$\begin{aligned} \mathfrak{A} = & X\left[-2A + x\left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz}\right) - \left(x\frac{dA}{dx} + y\frac{dA}{dy} + z\frac{dA}{dz}\right)\right] \\ & + Y\left[-2B + y\left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz}\right) - \left(x\frac{dB}{dx} + y\frac{dB}{dy} + z\frac{dB}{dz}\right)\right] \\ & + Z\left[-2C + z\left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz}\right) - \left(x\frac{dC}{dx} + y\frac{dC}{dy} + z\frac{dC}{dz}\right)\right], \end{aligned}$$

ou, en réduisant,

$$\begin{aligned} \mathfrak{A} = & -2(AX + BY + CZ) \\ & + (Xx + Yy + Zz)\left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz}\right) \\ & - (m-1)DT \\ & - T\left(x\frac{dA}{dt} + y\frac{dB}{dt} + z\frac{dC}{dt}\right) \\ & + t\left(X\frac{dA}{dt} + Y\frac{dB}{dt} + Z\frac{dC}{dt}\right). \end{aligned}$$

Puis un terme $\mathfrak{B} = \frac{dT}{dx^2}$, où

$$\begin{aligned} \mathfrak{B} = & -X\left[(Cx - Az)\frac{dT}{dx} - (Ay - Bx)\frac{dT}{dy}\right] \\ & -Y\left[(Ay - Bx)\frac{dT}{dx} - (Bz - Cy)\frac{dT}{dz}\right] \\ & -Z\left[(Bz - Cy)\frac{dT}{dy} - (Cx - Az)\frac{dT}{dx}\right], \end{aligned}$$

ou, en réduisant,

$$\begin{aligned}
 \mathfrak{B} &= \frac{dT}{dx} [(AX + BY + CZ)x - A(xX + yY + zZ)] \\
 &\quad + \frac{dT}{dy} [(AX + BY + CZ)y - B(xX + yY + zZ)] \\
 &\quad + \frac{dT}{dz} [(AX + BY + CZ)z - C(xX + yY + zZ)] \\
 &\quad - DT \left(x \frac{dT}{dx} + y \frac{dT}{dy} + z \frac{dT}{dz} \right) \\
 &\quad + tT \left(A \frac{dT}{dx} + B \frac{dT}{dy} + C \frac{dT}{dz} \right) \\
 &= -(m-1)DT^2 \\
 &\quad + Tt \left(A \frac{dT}{dx} + B \frac{dT}{dy} + C \frac{dT}{dz} \right).
 \end{aligned}$$

Donc, en réunissant les deux parties, $0 = \mathfrak{A} + \frac{\mathfrak{B}}{T}$, c'est-à-dire,

$$\begin{aligned}
 0 &= - \left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz} + \frac{dD}{dt} \right) \\
 &\quad + t \frac{d}{dt} (AX + BY + CZ + DT) \\
 &= -T \left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz} + \frac{dD}{dt} \right) \\
 &\quad + tQT,
 \end{aligned}$$

ou, en omettant le facteur tT , on obtient enfin

$$Q = - \left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz} + \frac{dD}{dt} \right),$$

c'est-à-dire que les fonctions A, B, C, D sont telles que

$$AX + BY + CZ + DT = - \left(\frac{dA}{dx} + \frac{dB}{dy} + \frac{dC}{dz} + \frac{dD}{dt} \right) f;$$

et, cela étant, l'expression générale $d\Omega$, et de même chacune des expressions $d\Omega_x, d\Omega_y, d\Omega_z, d\Omega_t$, sera égale à une différentielle totale.

Cambridge, le 8 Janvier 1886.

853. Note on a Formula for $\Delta^n 0^i / n^i$ when n, i are Very Large Numbers.

[From the *Proceedings of the Royal Society of Edinburgh*, t. xiv. (1887), pp. 149–153.]

THE following formula

$$\frac{\Delta^n 0^i}{n^i} = \left[1 + \left(\frac{i+1-2n}{2n} \right) q - \binom{n+i+2}{2} q^2 \dots \right], \quad q = e^{-(i+1)/n},$$

is given by Laplace (*Théorie Analytique des Probabilités*, 2nd ed., Paris, 1814, p. 195) as an approximate value of $\frac{\Delta^n 0^i}{n^i}$ when n and i are very large numbers, and is applied immediately afterwards to the case where i is of the order $n \log n$. As remarked by Professor Tait, it is certainly not applicable to the case where i is of the order n ; for taking $i = An$, where A is a given number however large, then q is indefinitely near to the very small value e^{-A} , but nevertheless the last term $-\frac{1}{2}(n+i+2)q^2$ by taking n sufficiently large, may be made as large as we please, and the value would thus come out negative. It is thus necessary that i should be at least of the order $n \log n$; but it may be of any higher order.

Writing for greater convenience $\frac{i+1}{n} = r$, $nq = re^{-1/n} r(1-X)$, if $X = 1 - e^{-1/n}$; and the formula becomes

$$\frac{\Delta^n 0^i}{n^i} = e^{-r(1-X)} \left[1 + \frac{i+1-2n}{2n^2} e^{-r/n} r - \frac{n+i+2}{2n^2} e^{-r/n} r^2 \dots \right].$$

Here $X = \frac{1}{n} - \frac{1}{1 \cdot 2} \frac{1}{n^2} + \frac{1}{1 \cdot 2 \cdot 3} \frac{1}{n^3} + \&c.$, and the exponential $e^{rX} = 1 + rX + \frac{r^2 X^2}{1 \cdot 2} + \dots$ is thus also expansible in negative powers of n ; the formula becomes

$$\frac{\Delta^n 0^i}{n^i} = e^{-r} \left(1 + rX + \frac{r^2 X^2}{1 \cdot 2} + \dots \right) \left[1 + \frac{i+1-2n}{2n^2} e^{-r/n} r - \frac{n+i+2}{2n^2} e^{-r/n} r^2 \dots \right];$$

viz. putting for X its value,

$$\begin{aligned} &= e^{-r} \left\{ 1 \right. \\ &\quad + r \left(\frac{i+1-2n}{2n^2} e^{-r/n} + 1 - e^{-r/n} \right) \\ &\quad + r^2 \left(\frac{-n-i-2}{2n^2} e^{-r/n} + \frac{i+1-2n}{2n^2} (1 - e^{-r/n}) e^{-r/n} + \frac{1}{2} (1 - e^{-r/n})^2 \right) \\ &\quad \left. + \&c. \right\}; \end{aligned}$$

or finally, expanding $e^{-r/n}$ and taking the whole result as far as $\frac{1}{n^2}$, the coefficient of r is

$$\left(-\frac{1}{n} + \frac{i+1}{n^2} \right) \left(1 - \frac{1}{n} \right) + \frac{1}{n} - \frac{1}{2n^2} = \frac{1 + \frac{1}{2}i}{n^2};$$

the coefficient of r^2 is

$$\left(-\frac{1}{n} + \frac{-2-i}{n^2} \right) \left(1 - \frac{1}{n} \right) + \left(-\frac{1}{n} + \frac{4i}{n^2} \right) \frac{1}{n} + \frac{1}{2n^2} = -\frac{1}{n} - \frac{1}{2} - \frac{1}{2}i;$$

whence the formula becomes

$$\frac{\Delta^n 0^i}{n^i} = e^{-r} \left\{ 1 + r \cdot \frac{1 + \frac{1}{2}i}{n^2} + \frac{r^2}{1 \cdot 2} \left(-\frac{1}{n} - \frac{-1-i}{n^2} \right) \dots \right\}.$$

It seems to me that the correct result up to this order of approximation is

$$\frac{\Delta^n 0^i}{n^i} = e^{-r} \left\{ 1 + r \cdot \frac{1 + \frac{1}{2}i}{n^2} + \frac{r^2}{1 \cdot 2} \left(-\frac{1}{n} + \frac{-1}{n^2} \right) \right\}.$$

My investigation is as follows: we have

$$\frac{\Delta^n 0^i}{n^i} = 1 - \frac{n}{1} \left(1 - \frac{1}{n}\right)^i + \frac{n \cdot n - 1}{1 \cdot 2} \left(1 - \frac{2}{n}\right)^i + \cdots,$$

the series being a finite one; but the number of terms is very large. But observe that, however large n is, we can take i so large that the second term $n \left(1 - \frac{1}{n}\right)^i$ may be as small as we please; taking this term to be of moderate amount, say $= r_1$, the subsequent terms will be not very different from $\frac{r_1^2}{1 \cdot 2}, \frac{r_1^3}{1 \cdot 2 \cdot 3}, \dots$, and the approximate value is $1 - r_1 + \frac{r_1^2}{1 \cdot 2} - \&c.$, which is a convergent series having its sum $= e^{-r_1}$.

To work this properly out, I represent the successive terms by $r_1, \frac{r_2}{1 \cdot 2}, \frac{r_3}{1 \cdot 2 \cdot 3}, \dots$, so that the series is

$$= 1 - r_1 + \frac{r_2}{1 \cdot 2} - \frac{r_3}{1 \cdot 2 \cdot 3} + \cdots.$$

Taking r a value at pleasure not very different from r_1 , and multiplying by

$$(1 + \nu) e^{-r}, \quad e^\nu = e^{-r} \cdot \left(1 + r + \frac{r^2}{1 \cdot 2} + \cdots\right),$$

the sum is

$$= e^{-r} \left\{ 1 + (r - r_1) + \frac{1}{1 \cdot 2} (r^2 - 2rr_1 + r_2) + \frac{1}{1 \cdot 2 \cdot 3} (r^3 - 3r^2r_1 + 3rr_2 - r_3) + \cdots \right\}.$$

Assuming now $r = ne^{-i/n}$, we have

$$r_1 = n \left(1 - \frac{1}{n}\right)^i = ne^{i \log(1-1/n)} = r(1 + X_1),$$

where $X_1 = e^{-\frac{i}{2} \frac{1}{n} - \frac{i}{3} \frac{1}{n^2} - \cdots} - 1$; and similarly

$$r_2 = n \cdot n - 1 \cdot \left(1 - \frac{2}{n}\right)^i = n^2 \left(1 - \frac{1}{n}\right)^i e^{i \log(1-2/n)} = \left(1 - \frac{1}{n}\right) r^2 (1 + X_2),$$

where $X_2 = e^{-\frac{2i}{2} \frac{1}{n} - \frac{2^2 i}{3} \frac{1}{n^2} - \cdots} - 1$; also

$$r_3 = n \cdot n - 1 \cdot \left(1 - \frac{3}{n}\right)^i = \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) r^3 (1 + X_3),$$

where $X_3 = e^{-\frac{3i}{2} \frac{1}{n} - \frac{3^2 i}{3} \frac{1}{n^2} - \cdots}$, and so on. It is now easy to calculate the successive terms $r - r_1, r^2 - 2rr_1 + r_2$, &c.; and it is to be observed that, in the parts independent of the X 's, we have only terms divided by n, n^2 , or higher powers of n : thus in $r^4 - 4r^3r_1 + 6r^2r_2 - 4rr_3 + r_4$, we have r^4 multiplied by

$$1 - 4 + 6 \left(1 - \frac{1}{n}\right) - 4 \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \left(1 - \frac{3}{n}\right) = \frac{3}{n^2} - \frac{6}{n^3}.$$

We thus obtain the formula

$$\frac{\Delta^n 0^i}{n^i} = e^{-r} \left\{ 1 + r(-1 X_1) + \frac{r^2}{1 \cdot 2} \left[-\frac{1}{n} - 2X_1 + \left(1 - \frac{1}{n}\right)X_2\right] + \frac{r^3}{1 \cdot 2 \cdot 3} \left[-\frac{2}{n^2} - 3X_1 + 3\left(1 - \frac{1}{n}\right)X_2 - \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)X_3\right] + \cdots \right\},$$

with the next term

$$+ \frac{r^4}{1 \cdot 2 \cdot 3 \cdot 4} \left[-\frac{3}{n^2} - \frac{6}{n^3} - 4X_1 + 6\left(1 - \frac{1}{n}\right)X_2 - 4\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)X_3 + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\left(1 - \frac{3}{n}\right)X_4\right] + \cdots,$$

where $r = ne^{-i/n}$ as above, and $X_1, X_2, \dots$ have the above-mentioned values, the exponentials being expanded in negative powers of n .

Writing

$$X_1 = -\frac{1}{2}\frac{i}{n^2}, \quad X_2 = -\frac{2i}{n^2},$$

we have

$$\frac{\Delta^n 0^i}{n^i} = e^{-r} \left\{ 1 + r \cdot \frac{1 + \frac{1}{2}i}{n^2} + \frac{r^2}{2} \left(-\frac{1}{n} + \frac{-i}{n^2} \right) \right\},$$

which is the foregoing approximate value.

854. An Algebraical Transformation.

[From the *Messenger of Mathematics*, vol. xv. (1886), pp. 58, 59.]

THE following algebraical transformation occurs in a paper by Hermite “On the theory of the Modular Equations,” *Comptes Rendus*, t. XLVIII. (1859), p. 1100.

Writing $q = 1 - 2u^4$, $l = 1 - 2v^4$, then in the transformation of the fifth order, the modular equation was expressed by Jacobi in the form

$$\Omega = (q - l)^6 - 256(1 - q^2)(1 - l^2) \left[16ql((9 - q^2l) + 9(45 - ql)(q - l))^2 \right] = 0;$$

and if we write herein $q = 1 - 2x$, $l = \frac{x+1}{x-1}$, or, what is the same thing, establish between q , l the relation $q = \frac{1+ql}{l-1}$, that is, between u , v the relation $v^4 = 1 \div (1 - u^4)$, then the function Ω becomes

$$\Omega = \frac{64}{(1-x)^6} \left[(x^2 - x + 1)^4 + 64(1-x)^2 x^2 \{ (x^2 - x + 1)^3 + 2^4(x^2 - x + 1)^2 + 3^4(x^2 - x + 1) \} \right];$$

or, what is the same thing, the equation $\Omega = 0$ gives for $\frac{x^2 - x + 1}{(x^2 - x)^2}$ the values -2^4 and -3^4 .

We, in fact, have

$$q - l = 3 + ql = -\frac{2(x^2 - x + 1)}{1 - x}, \quad 1 - q^2 = 4x(1 - x), \quad 1 - l^2 = -\frac{4x}{(1 - x)^2},$$

and therefore

$$(1 - q^2)(1 - l^2) = -\frac{16x^2}{1 - x}.$$

Hence

$$\Omega = \frac{64}{(1-x)^6} \left[(x^2 - x + 1)^6 + 64(1-x)^2 x^2 \left\{ 16ql((9 - q^2l) + 9(45 - ql)(3 - 2ql))^2 \right\} \right];$$

and, putting for a moment $ql = \theta - 3$, the term in $\{ \quad \}$ is found to be

$$= 7l^4\theta + 34560\theta^3 - 6912;$$

viz. this is

$$\begin{aligned} &= \frac{56(x^2 - x + 1)^6}{(1-x)^6} + \frac{6912(x^2 - x + 1)}{1-x} - 6912, \\ &= \frac{8}{(1-x)^6} [7(x^2 - x + 1)^6 + 864(x-1)^2(x^2 - x + 1) + 864(x-1)^4], \\ &= \frac{8}{(1-x)^6} [7(x^2 - x + 1)^6 + 864(x^2 - x)^2]. \end{aligned}$$

Hence

$$\Omega = \frac{64}{(1-x)^6} [(x^2 - x + 1)^6 + 512(x^2 - x)^2 \{ 7(x^2 - x + 1)^6 + 864(x^2 - x)^2 \}],$$

which is

$$= \frac{64}{(1-x)^6} [(x^2 - x + 1)^6 + 2^4(x^2 - x)^2] [(x^2 - x + 1)^6 + 2^4 \cdot 3^4(x^2 - x)^2].$$

855. Solution of $(a, b, c, d) = (a^2, b^2, c^2, d^2)$.

[From the *Messenger of Mathematics*, vol. xv. (1886), pp. 59–61.]

It is required to find four quantities (no one of them zero) which are in some order or other equal to their squares, say

$$(a, b, c, d) = (a^2, b^2, c^2, d^2).$$

Supposing that the required quantities (a, b, c, d) are the roots of the biquadratic equation

$$x^4 + px^3 + qx^2 + rx + s = 0,$$

($s \neq 0$), then the function

$$(x^4 + px^3 + qx^2 + rx + s)^2 \text{ must be } = x^8 + px^6 + qx^4 + rx^2 + s;$$

and we have thus the conditions

$$p^2 - 2q = p, \quad 2s + 2pr - q^2 = q, \quad 2qs - r^2 = r, \quad s^2 = s,$$

the last of which (since s is not $= 0$) gives $s = 1$, and the others then become

$$2q = p^2 + p, \quad 2(pr - 1) = q^2 - q, \quad 2q = r^2 + r;$$

viz. regarding p, q, r as the coordinates of a point in space, this is determined as the intersection of three quadric surfaces, and the number of solutions is thus $= 8$.

We, in fact, have $2q = p^2 + p = r^2 + r$; that is, $p^2 + p = r^2 + r$, or $(p - r)(p + r + 1) = 0$; hence $r = p$ or $r = -1 - p$.

First, if $r = p$: here $2q = p^2 + p$, $2(p^2 - 1) = q^2 - q$: the last equation multiplied by 4 gives

$$8(p^2 - 1) = (p^2 + p)(p^2 + p - 2),$$

that is, $p^4 + 2p^3 - 8 = 0$ or $(p - 1)(p^2 + 3p + 8) = 0$.

If $p^4 - 1 = 0$, then either $p = 1$, giving $q = 1$, $r = 1$, and hence the equation is

$$x^4 + x^3 + x^2 + x + 1 = 0,$$

or else $p = -1$, giving $q = 0$, $r = -1$, and hence the equation is

$$x^4 - x^3 + 1 = 0, \text{ that is, } (x - 1)(x^3 - x + 1) = 0.$$

If $p^2 + 2p - 8 = 0$, then either $p = 2$, giving $q = 3$, $r = 2$, and hence the equation is

$$x^4 + 2x^3 + 3x^2 + 2x + 1 = 0, \text{ that is, } (x^2 + x + 1)^2 = 0,$$

or else $p = -4$, giving $q = 6$, $r = -4$, and hence the equation is $x^4 + 6x^3 - 4x + 1 = 0$, that is, $(x - 1)^4 = 0$.

Secondly, if $r = -1 - p$; here

$$2q = p^2 + p, \quad 2(-p^2 - p - 1) = q^2 - q;$$

the last equation multiplied by 4 gives

$$8(-p^2 - p - 1) = (p^2 + p)(p^2 + p - 2),$$

that is,

$$p^4 + 2p^3 + 7p^2 + 6p + 8 = 0, \text{ or } (p^2 + p + 4)(p^2 + p + 2) = 0.$$

If $p^2 + p + 4 = 0$, then $p = \frac{1}{2}\{-1 \pm i\sqrt{15}\}$, whence

$$r = \frac{1}{2}\{-1 \mp i\sqrt{15}\}, \quad 2q = p^2 + p = -4, \text{ or } q = -2;$$

and the equation is

$$x^4 + \frac{1}{2}\left\{-1 \pm i\sqrt{15}\right\}x^3 - 2x^2 + \frac{1}{2}\left\{-1 \mp i\sqrt{15}\right\}x + 1 = 0.$$

If $p^2 + p + 2 = 0$, then $p = \frac{1}{2}\{-1 \pm i\sqrt{7}\}$, whence

$$r = \frac{1}{2}\left\{-1 \mp i\sqrt{7}\right\}, \quad 2q = p^2 + p = -2, \text{ or } q = -1;$$

and the equation is

$$x^4 + \frac{1}{2}\left\{-1 \pm i\sqrt{7}\right\}x^3 - x^2 + \frac{1}{2}\left\{-1 \mp i\sqrt{7}\right\}x - 1 = 0,$$

that is,

$$(x-1)\left[x^3 + \frac{1}{2}\left\{1 \pm i\sqrt{7}\right\}x^2 + \frac{1}{2}\left\{-1 \mp i\sqrt{7}\right\}x - 1\right] = 0.$$

We thus see that the eight equations are

$$\begin{aligned} 1 \quad & (x-1)^4 = 0, \\ 1 \quad & (x^2 + x + 1)^2 = 0, \\ 1 \quad & (x-1)(x^3 - x + 1) = 0, \\ 1 \quad & x^4 + x^3 + x^2 + x + 1 = 0, \\ 2 \quad & (x-1)\left[x^3 + \frac{1}{2}(1 \pm i\sqrt{7})x^2 + \frac{1}{2}(-1 \mp i\sqrt{7})x - 1\right] = 0, \\ 2 \quad & x^4 + \frac{1}{2}(-1 \pm i\sqrt{15})x^3 - 2x^2 + \frac{1}{2}(-1 \mp i\sqrt{15})x + 1 = 0, \end{aligned}$$

(totalling 8) and it hence appears that, writing γ, ϵ, θ to denote respectively an imaginary cube root, fifth root, and seventh root of unity, then the values of (a, b, c, d) are

$$\begin{aligned} & 1, 1, 1, 1; \\ & \gamma, \gamma, \gamma^2, \gamma^2; \\ & 1, 1, \gamma, \gamma^2; \\ & \epsilon, \epsilon^2, \epsilon^3, \epsilon^4; \\ & \epsilon^2\gamma, \epsilon\gamma^2, \epsilon^4\gamma, \epsilon^3\gamma^2; \\ & \epsilon^2\gamma, \epsilon\gamma^2, \epsilon^3\gamma^2, \epsilon^4\gamma; \\ & 1, \theta, \theta^2, \theta^4; \\ & 1, \theta^3, \theta^6, \theta^5; \end{aligned}$$

viz. for each of these systems we have the required relation

$$(a, b, c, d) = (a^2, b^2, c^2, d^2).$$

It may be noticed that out of the eight equations we have the following three which are irreducible:

$$\begin{aligned} & x^4 + x^3 + x^2 + x + 1 = 0, \\ & x^4 + \frac{1}{2}(-1 + i\sqrt{15})x^3 - 2x^2 + \frac{1}{2}(-1 - i\sqrt{15})x + 1 = 0, \\ & x^4 + \frac{1}{2}(-1 - i\sqrt{15})x^3 - 2x^2 + \frac{1}{2}(-1 + i\sqrt{15})x + 1 = 0. \end{aligned}$$

Each of these is an Abelian equation, viz. the roots are of the form

$$\alpha, \theta(\alpha), \theta^2(\alpha), \theta^3(\alpha), \quad (= \alpha, \alpha^a, \alpha^{a^2}, \alpha^{a^3}),$$

where $\theta^4(\alpha) = \alpha$, not identically but in virtue of the value of α , viz. we have $\alpha^{a^4} = \alpha^{16} = \alpha$, $\alpha^{15} = 1$ (in the first equation $a^5 = 1$, and therefore $a^{15} = 1$; in each of the other two, a^{15} is the lowest power which is $= 1$).

In the first equation, we have evidently

$$x^4 + x^3 + x^2 + x + 1$$

as the irreducible factor of $x^5 - 1$.

The second and third equations combined together give

$$\left(x^4 - \frac{1}{2}x^3 + \frac{1}{2}x - 2x^2 + \frac{15}{4}\right)(x^4 - x^3) = 0;$$

that is,

$$x^8 - x^7 + x^6 - x^4 + x^2 - x + 1 = 0,$$

where the left-hand side is the irreducible factor of $x^{15} - 1$.

856. Note on a Cubic Equation.

[From the *Messenger of Mathematics*, vol. xv. (1886), pp. 62–64.]

CONSIDER the cubic equation

$$x^3 + 3cx + d = 0;$$

then effecting upon this the Tschirnhausen-Hermite transformation

$$y = xT_1 + (x^2 + 2c)T_2,$$

the resulting equation in y is

$$\begin{aligned} y^3 + 3y(cT_1^2 + dT_1T_2 - c^2T_2^2) \\ + dT_1^3 - 6c^2T_1^2T_2 - 3cdT_1T_2^2 - (d^2 + 2c^3)T_2^3 = 0, \end{aligned}$$

and this will be

$$y^3 + 3cy + d = 0,$$

if only

$$c = cT_1^2 + dT_1T_2 - c^2T_2^2, \quad d = dT_1^3 - 6c^2T_1^2T_2 - 3cdT_1T_2^2 - (d^2 + 2c^3)T_2^3,$$

equations which give

$$(d^2 + 4c^3) = (d^2 + 4c^3)(T_1^3 + 3cT_1T_2^2 + dT_2^3),$$

viz. assuming that $d^2 + 4c^3$ not = 0, this is

$$1 = T_1^3 + 3cT_1T_2^2 + dT_2^3.$$

Hence the coefficients T_1, T_2 being such as to satisfy these relations, the equation in y is identical with the equation in x ; or, what is the same thing, if α, β, γ are the roots of the equation in x , then we have between these roots the relations

$$\beta = \alpha T_1 + (\alpha^2 + 2c)T_2,$$

$$\gamma = \beta T_1 + (\beta^2 + 2c)T_2,$$

$$\alpha = \gamma T_1 + (\gamma^2 + 2c)T_2,$$

viz. the general cubic equation $x^3 + 3cx + d = 0$, adjoining thereto the radicals T_1, T_2 , may be regarded as an Abelian equation.

In particular, if $c, d = -1, 1$, then we may write $T_1 = 0, T_2 = 1$; the cubic equation is here

$$x^3 - 3x + 1 = 0,$$

and the roots α, β, γ are such that $\beta = \alpha^2 - 2, \gamma = \beta^2 - 2, \alpha = \gamma^2 - 2$; in fact, taking θ a primitive ninth root of unity, $\theta^6 + \theta^3 + 1 = 0$; we have $\alpha, \beta, \gamma = \theta + \theta^8, \theta^2 + \theta^7, \theta^4 + \theta^5$, values which satisfy $x^3 - 3x + 1 = 0$, and the relations in question.

The same question may be considered from a different point of view. Take the transforming equation to be

$$y = A + Bx + Cx^2,$$

then assuming that the values of y corresponding to the values $x = \alpha, \beta, \gamma$ are β, γ, α respectively, we have

$$\beta = A + B\alpha + C\alpha^2,$$

$$\gamma = A + B\beta + C\beta^2,$$

$$\alpha = A + B\gamma + C\gamma^2,$$

and the transforming equation thus is

$$\begin{vmatrix} y, & 1, & x, & x^2 \\ \beta, & 1, & \alpha, & \alpha^2 \\ \gamma, & 1, & \beta, & \beta^2 \\ \alpha, & 1, & \gamma, & \gamma^2 \end{vmatrix} = 0.$$

This may also be written

$$\begin{aligned} & (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta) \left\{ y - \frac{1}{3}(\alpha + \beta + \gamma + x) \right\} \\ &= \beta^2\gamma + \gamma^2\alpha + \alpha^2\beta - \frac{1}{3}(\beta^3 + \gamma^3 + \alpha^3 + \beta^2\gamma + \gamma^2\alpha + \alpha^2\beta + \alpha\beta^2 + \beta\gamma^2 + \gamma\alpha^2) \\ & \quad + x \left\{ \beta\gamma + \gamma\alpha + \alpha\beta - \frac{1}{3}(\alpha^2 + \beta^2 + \gamma^2) \right\}. \end{aligned}$$

We have

$$(\beta - \gamma)^2(\gamma - \alpha)^2(\alpha - \beta)^2 = -27(a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd) = -27\Delta,$$

or, say

$$(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta) = \frac{3(\omega - \omega^2)}{a^2} \sqrt{(\Delta)},$$

if Δ be the discriminant, and ω an imaginary cube root of unity, $\{(\omega - \omega^2)^2 = -3\}$.

The remaining functions of α, β, γ are of course expressible rationally in terms of the coefficients: we have

$$\begin{aligned} \Sigma\beta^2\gamma &= \frac{1}{a^3}(-6bd + 9c^2), \\ \Sigma\beta\gamma &= \frac{1}{a^3}(-3abd - 18ac^2 + 27b^2c), \\ \Sigma\alpha^2 &= \frac{1}{a^3}(-3a^2d + 7abc - 9b^3), \\ \Sigma\beta\gamma &= \frac{1}{a^2}(3ad - 9bc), \\ \Sigma\alpha\beta &= \frac{3c}{a}, \\ \Sigma\alpha^2 &= \frac{1}{a^2}(9b^2 - 6ac), \end{aligned}$$

and the final result is

$$\begin{aligned} \frac{1}{2}(\omega - \omega^2)\sqrt{(\Delta)}\{2a(y + x) + 3b\} &= -abd + 4ac^2 - 3b^2c \\ & \quad + x(-a^2d + 7abc - 6b^3) \\ & \quad + x^2(2a^2c - 2ab^2); \end{aligned}$$

viz. we have thus an automorphic transformation of the equation

$$ax^3 + 3bx^2 + 3cx + d = 0.$$

857. Analytical Geometrical Note on the Conic.

[From the *Messenger of Mathematics*, vol. xv. (1886), p. 192.]

TAKE (X, Y, Z) the coordinates of a point on the conic $yz + zx + xy = 0$, so that $YZ + ZX + XY = 0$; clearly (Y, Z, X) and (Z, X, Y) are the coordinates of two other points on the same conic; I say that the three points are the vertices of a triangle circumscribed about the conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0.$$

In fact, the equation of one of the sides is

$$\begin{vmatrix} x, & y, & z \\ X, & Y, & Z \\ Y, & Z, & X \end{vmatrix} = 0,$$

say this is $AX + BY + CZ = 0$, where $A, B, C = XY - Z^2, YZ - X^2, XZ - Y^2$; and the condition in order that this side may touch the conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0$$

is

$$BC + CA + AB = 0.$$

But we have

$$\begin{aligned} BC + CA + AB &= Y^2Z^2 + Z^2X^2 + X^2Y^2 - X(Y^3 + Z^3) - Y(Z^3 + X^3) - Z(X^3 + Y^3) \\ &\quad + XYZ + XYZ + XYZ \\ &= -(YZ + ZX + XY)(-X^3 - Y^3 - Z^3 + YZX + ZXY + XYZ); \end{aligned}$$

and similarly for the other two sides. The point (X, Y, Z) is an arbitrary point on the conic $yz + zx + xy = 0$; and we thus see that we have a singly infinite series of triangles each inscribed in this conic and circumscribed about the conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0.$$

858. Comparison of the Weierstrassian and Jacobian Elliptic Functions.

[From the *Messenger of Mathematics*, vol. xvi. (1887), pp. 129–132.]

THE Weierstrassian function $\sigma(u)$ corresponds of course with Jacobi's $H(u)$, but it is worth while to establish the actual formulas of transformation.

Writing, for a moment,

$$\omega = \omega_1 + i\omega_2, \quad \omega' = \omega'_1 + i\omega'_2,$$

it is convenient to assume $\omega_1\omega'_2 - \omega_2\omega'_1$ positive; we then have

$$2(\eta\omega' - \eta'\omega) = \pi i;$$

in particular, this will be the case if $\omega = \omega_1$, $\omega' = i\omega_2$, where ω_1, ω_2 are each positive.

To reduce the periods into the Jacobian form, we may assume

$$\omega = \lambda K, \quad \omega' = \lambda i K',$$

(where observe that, if as usual k, K, K' are each real and positive, and if as above $\omega = \omega_1$, $\omega' = i\omega_2$, ω_1 and ω_2 positive, then also λ will be real and positive). We have

$$\frac{iK'}{K} = \frac{\omega'}{\omega}, \quad \text{or say} \quad q = e^{-\frac{\pi K'}{K}} = e^{\frac{i\pi\omega'}{\omega}},$$

which determines first q , and thence k, K, K' as functions of $\frac{\omega'}{\omega}$, and we then have

$$\lambda = \frac{\omega}{K} = \frac{-i\omega'}{K'},$$

either of which equations gives λ as a function of ω, ω' . Conversely, starting with k, λ , the original equations give the values of ω, ω' ; those of η, η' will be determined presently.

The form of relation is at once seen to be

$$H(u) = Ae^{Bu^2}\sigma(\lambda u),$$

and observing that, for u small, we have $H(u) = \sqrt{\left(\frac{2kK}{\pi}\right)}u$ and $\sigma(\lambda u) = \lambda u$, we have $A = \frac{1}{\lambda}\sqrt{\left(\frac{2kK}{\pi}\right)}$;

I first write down and afterwards verify the value of B , viz. this is $-\frac{1}{2}\frac{\lambda\eta}{\omega}$; and the formula thus is

$$H(u) = \frac{1}{\lambda}\sqrt{\left(\frac{2kK}{\pi}\right)}e^{-\frac{1}{2}\frac{\lambda\eta}{\omega}u^2}\sigma(\lambda u).$$

In fact, for u writing successively $u + 2K$, and $u + 2iK'$, we obtain

$$\frac{H(u + 2K)}{H(u)} = e^{-\frac{2\lambda\eta}{\omega}K(u+K)} \frac{\sigma(\lambda u + 2\omega)}{\sigma(\lambda u)},$$

$$\frac{H(u + 2iK')}{H(u)} = e^{-\frac{2\lambda\eta}{\omega}iK'(u+iK')} \frac{\sigma(\lambda u + 2\omega')}{\sigma(\lambda u)},$$

which should be satisfied in virtue of

$$\frac{H(u + 2K)}{H(u)} = -1, \quad \frac{\sigma(\lambda u + 2\omega)}{\sigma(\lambda u)} = -e^{2\eta(\lambda u + \omega)},$$

$$\frac{H(u + 2iK')}{H(u)} = e^{-\frac{\pi}{K}(u+iK')}, \quad \frac{\sigma(\lambda u + 2\omega')}{\sigma(\lambda u)} = -e^{2\eta'(\lambda u + \omega')};$$

viz. we ought to have

$$0 = \frac{\lambda\eta}{\omega} 2K(u + K) + 2\eta(\lambda u + \omega)$$

$$-\frac{i\pi}{K}(u + iK') = \frac{\lambda\eta}{\omega} 2iK'(u + iK') + 2\eta'(\lambda u + \omega').$$

The first of these is

$$0 = -\frac{\lambda K}{\omega}(u + K) + \left(\lambda + \frac{\omega}{\lambda}\right),$$

that is,

$$0 = (-1 + 1)(u + K);$$

and the second is

$$0 = \left(\frac{\lambda iK'}{\lambda K} - \frac{\lambda iK'}{\omega}\right)(u + iK') + \eta'\left(u + \frac{\omega'}{\lambda}\right),$$

viz. for iv' writing $2(\eta\omega' - \eta'\omega)$, this is

$$0 = \left(\frac{\eta\omega' - \eta'\omega}{\omega} - \frac{\eta\omega'}{\omega} + \eta'\right)(u + iK'),$$

and the two equations are thus each of them an identity.

The Weierstrassian function $\wp(u)$ is defined as

$$\wp(u) = -\frac{d^2}{du^2} \log \sigma(u),$$

or, what is the same thing, we have

$$\wp(u) = -\frac{d}{du} \frac{\sigma'(u)}{\sigma(u)}.$$

Hence

$$\wp(\lambda u) = -\frac{1}{\lambda} \frac{d}{du} \frac{\sigma'(\lambda u)}{\sigma(\lambda u)}.$$

But

$$\frac{H'(u)}{H(u)} = \frac{\lambda^2 \eta u}{\omega} + \frac{\lambda \sigma'(\lambda u)}{\sigma(\lambda u)},$$

or

$$\frac{1}{\lambda} \frac{d}{du} \frac{H'(u)}{H(u)} = -\eta + \wp(\lambda u).$$

But from the equation $\sqrt{k} \operatorname{sn} u = \frac{H(u)}{\Theta(u)}$, we obtain

$$\frac{d}{du} \frac{H'(u)}{H(u)} = Z'(u) - \frac{1}{\operatorname{sn}^2 u} + k^2 \operatorname{sn}^2 u, \quad = 1 - \frac{E}{K} - \frac{1}{\operatorname{sn}^2 u},$$

and consequently

$$\wp(\lambda u) = \frac{\eta}{\omega} - \frac{1}{\lambda^2} \left(1 - \frac{E}{K}\right) + \frac{1}{\lambda^2 \operatorname{sn}^2 u},$$

where, on the right-hand side, expanding in ascending powers of u , the constant term is

$$= -\frac{\eta}{\omega} - \frac{1}{\lambda^2} \left(1 - \frac{E}{K}\right) + \frac{1}{\lambda^2} \cdot \frac{1}{3} (1 + k^2).$$

But in the function $\wp(\lambda u)$ this constant term is $= 0$, and we thus have

$$\frac{\eta}{\omega} = \frac{1}{\lambda^2} \left\{ \frac{1}{3} (1 + k^2) - 1 + \frac{E}{K} \right\};$$

and then, since

$$\frac{\eta}{\omega} - \frac{\eta'}{\omega'} = \frac{\frac{1}{2}\pi i}{\omega\omega'} = \frac{\frac{1}{2}\pi i}{\lambda^2 i K K'},$$

we have

$$\frac{\eta'}{\omega'} = -\frac{\frac{1}{2}\pi i}{\lambda^2 i K K'} + \frac{1}{\lambda^2} \left\{ \frac{1}{3} (1 + k^2) - 1 + \frac{E}{K} \right\};$$

or, as these equations may also be written,

$$\eta = \frac{K}{\lambda} \left\{ \frac{1}{3}(1 + k^2) - 1 + \frac{E}{K} \right\},$$

$$\eta' = \frac{\frac{1}{2}\pi i}{\lambda K} + \frac{2iK'}{\lambda} \left\{ \frac{1}{3}(1 + k^2) - 1 + \frac{E}{K} \right\},$$

which are the values of η , η' . And we then have

$$\wp(\lambda u) = \frac{1}{\lambda^2} \frac{1}{3}(1 + k^2) + \frac{1}{\lambda^2 \operatorname{sn}^2 u},$$

the equation connecting $\wp(\lambda u)$ and $\operatorname{sn} u$.

859. On the Complex of Lines which meet a Unicursal Quartic Curve.

[From the *Proceedings of the London Mathematical Society*, vol. xvii. (1886), pp. 232–238.]

THE curve is taken to be that determined by the equations

$$x : y : z : w = 1 : \theta : \theta^2 : \theta^4,$$

viz. it is the common intersection of the quadric surface $\Theta = 0$, and the cubic surfaces $P = 0$, $Q = 0$, $R = 0$, where

$$\Theta = xw - yz, \quad P = x^2z - y^2w, \quad Q = xz^2 - y^2w, \quad R = xz - y^2.$$

Writing (a, b, c, f, g, h) as the six coordinates of a line, viz.

$$(a, b, c, f, g, h) = (\beta z - \gamma y, \gamma x - \alpha z, \alpha y - \beta x, \alpha w - \delta x, \beta w - \delta y, \gamma w - \delta z),$$

if $(\alpha, \beta, \gamma, \delta)$, (x, y, z, w) are the coordinates of any two points on the line; then, if the line meet the curve, we have

$$\begin{aligned} -h\theta + g\theta^2 + a\theta^4 &= 0, \\ -h + f\theta^2 + b\theta^4 &= 0, \\ g - f\theta + c\theta^4 &= 0, \\ -a - b\theta - c\theta^3 &= 0, \end{aligned}$$

from which four equations (equivalent, in virtue of the identity $af + bg + ch = 0$, to two independent equations), eliminating θ , we have the equation of the complex. The form may, of course, be modified at pleasure by means of the identity just referred to, but one form is

$$\Omega = a^4 - b^3h + bf^2g + cg^3 - acfh + 2c^2h^2 - 4a^2ch + af^3 - a^3f = 0,$$

as may be verified by substituting therein the values $a = \beta w - c\theta^3$, $g = f\theta - c\theta^2$, $h = f\theta^3 + b\theta^4$. The last-mentioned equation is thus the equation of the complex in question, in terms of the six coordinates (a, b, c, f, g, h) .

If for the six coordinates we substitute their values, $\beta z - \gamma y$, &c., we obtain

$$\Omega = (x, y, z, w)^4 (\alpha, \beta, \gamma, \delta)^4 = 0,$$

which, regarded as an equation in (x, y, z, w) , is the equation of the cone, vertex $(\alpha, \beta, \gamma, \delta)$, passing through the quartic curve; this equation should evidently be satisfied if only Θ, P, Q, R are each $= 0$, viz. must be a linear function of (Θ, P, Q, R) ; and by symmetry, it must be also a linear function of (Θ', P', Q', R') , where

$$\Theta' = \alpha\delta - \beta\gamma, \quad P' = \alpha^2\gamma - \beta^3, \quad Q' = \alpha\gamma^2 - \beta^2\delta, \quad R' = \alpha\gamma - \beta^2,$$

viz. the form is $\Omega = (\Theta', P', Q', R') (\Theta, P, Q, R)$, an expression with coefficients which are of the first or second degree in (x, y, z, w) and also of the first or second degree in $(\alpha, \beta, \gamma, \delta)$.

To work this out, I first arrange in powers and products of (α, δ) , (β, γ) , expressing the quartic functions of (x, y, z, w) in terms of (Θ, P, Q, R) , as follows:

$\Omega =$ expansion table

	a^4	$-b^3h$	$+bf^2g$	$+cg^3$	$-acfh$	$+2c^2h^2$	$-4a^2ch$	$+af^3$	$-a^3f$	sum	contribution
α^4										0	0
$\alpha^3\delta$		$-z^4$	$+yzw^2$	$-y^4$		$+2y^2z^2$				$-z^4 + yzw^2$	$-zR$
$\alpha^2\delta^2$			$-2xyzw$							$-2xyzw + 2y^2z^2$	$-2yz\Theta$
$\alpha\delta^3$			$+x^2yz$							$+x^2yz - y^4$	$+yP$
δ^4										0	0
$\alpha^3\beta$			$-zw^3$				$+zw^3$			0	0
$\alpha^2\beta\delta$			$+2xzw^2$		$+yz^2w$		$-3xzw^2$			$-xzw^2 + yz^2w$	$-zw\Theta$
$\alpha\beta\delta^2$			$-x^2zw$	$+3y^3w$	$-xyf^2w$	$-4xyz^2$	$+3x^2zw$			$+2x^2zw + 3y^3w - 3xyf^2w - 5xyz^2$	$+2xz\Theta - 3yQ$
$\beta\delta^3$				$+xy^3$			$-x^2y$			$+xy^3 - x^3z$	$-xP$
$\alpha^2\gamma$			$+z^3w$		$-yzw^2$		$-yw^3$			$+z^3w - yw^3$	$+wR$
$\alpha^2\gamma\delta$			$+3xz^3$		$-xyf^2w$		$-3xz^2w$			$+3xz^3 + 2xyw^2 - 5yzw^2$	$+2yw\Theta + 3zQ$
$\alpha\gamma\delta^2$			$+2x^2zw$	$-x^2y$	$+x^2z^2$		$-3x^2yw$	$+3xyw^2$		$+x^3y$	$-xy\Theta$
$\gamma\delta^3$				$-x^3y$				$+x^3y$		0	0
$\alpha^2\beta\delta^2$		$+xw^4$								$+xw^4 - yzw^3$	$+w^2\Theta$
$\alpha^2\beta\gamma$		$-3xz^2w$	$+xw^3$		$-yzw^2$	$+2y^2zw$	$+4yz^3$			$-3xz^2w + 3y^2zw - 2yzw^2 + 4yz^3 + 3xz\Theta$	$-3wQ$
$\alpha^2\gamma^2$		$-2c^2w^2$		$-3y^3w^2$	$-x^2w$	$+2xyzw$	$+8xyz^2$			$-3y^3w^2 - xz^2w + 2xyzw + 8xyz^2 - 3xz^3 + 10xyzw - 9y^2z^2$	$-4z^2\Theta + 3wQ$
$\alpha\beta^2\delta$		$-3xz^2$	$-2x^2w^2$		$+2xyzw$	$+8xyz^2$	$-9y^2z^2$			$-3xz^2 - 2x^2w^2 + 2xyzw + 8xyz^2 - 3xz^3 + 10xyzw - 9y^2z^2$	$-4y^2\Theta - 3xQ$
$\alpha\beta\gamma\delta$			$+x^2w$	$-2xyf^2w$	$+xy^2$		$-3xy^2w$			$-3xz^2w + 2xyf^2w + xz^2y + x^2y^2$	$+3xQ$
$\alpha\gamma^2\delta$		$+x^2w$	$-2xz^2$	$+x^3z$						$+x^3z - x^2y$	$+x^2\Theta$
$\beta\gamma\delta^2$				$+yw^3$	$+xzw^2$	$-4xyzw$	$-3yz^2w$			$+yw^3 - z^2w - xzw\Theta$	$-wR$
$\gamma^2\delta^2$				$+xww^2$	$-xyzw$	$+8yz^2w$				$+xzw - yz^2w$	$+zw\Theta$
$\alpha\beta^3$			$+yw^3$		$+xzw^2$	$-4xyzw$	$-4y^2z$		$-y^3z$	$+ywR\Theta - 5y^2zw$	$-5yw\Theta$
$\alpha\beta^2\gamma$			$+xww^3$	$-4y^2w^2$	$+8y^2zw$					$+3y^2zw - 5xy^2z + 5y^2zw$	$-3xz\Theta$
$\alpha\beta\gamma^2$			$+3x^2zw$	$-xy^2w$	$-4xyzw$	$+8xyz^2$	$-xyf^2w$			$+x^2yw - xy^2z$	$-5xz\Theta$
$\alpha\gamma^3$			$+3x^2zw$	$-xy^2w$	$-xw^3$	$-4xyf^2z$				$+x^2zw - xy^2z$	$+xz\Theta$
$\beta^3\delta$			$+x^2z$							$+x^3z - xy^3$	$+xP$
$\beta^2\gamma\delta$			$+3x^2z$	$+3xyw^2$						$+z^4 - xw^3$	$+zR - w^2\Theta$
$\beta\gamma^2\delta$					$+x^2yw$	$-x^2zw$	$-4x^2z^2$			$+2x^2w^2 - 8xyzw + 6y^2z^2$	$+(2xw - 6yz)\Theta$
$\gamma^3\delta$					$+x^2yw$	$+2x^2w^2$	$-8xyzw$			$-4y^3z + 4xy^2z$	$+4y^2\Theta$
β^4	$+z^4$						$-4y^3z$			$+z^4 - xw^3$	$+zR - w^2\Theta$
$\beta^3\gamma$	$-4yz^3$									$+4xz^2w$	$+4z^2\Theta$
$\beta^2\gamma^2$	$+6y^2z^2$				$+2x^2w^2$	$-8xyzw$	$+6y^2z^2$			$+2x^2w^2 - 8xyzw + 6y^2z^2$	$+(2xw - 6yz)\Theta$
$\beta\gamma^3$	$-4y^3z$									$+4y^2w$	$+4y^2\Theta$
γ^4	$+y^4$	$-x^3w$								$+y^4 - x^3w$	$-yP - x^2\Theta$

[Table reproduced from the scan: rows index the monomials of degree four in $(\alpha, \beta, \gamma, \delta)$, columns enumerate the monomials of Ω in the six line-coordinates (a, b, c, f, g, h) ; the penultimate column is the sum of the entries on that row, and the last column expresses that sum in the basis (Θ, P, Q, R) .]

Collecting the terms multiplied by P , Q , R , Θ , respectively, we have

$$\begin{aligned}\Omega = & P\{y\alpha\delta^3 - x\beta\delta^3 + 3w\alpha y^3 + xy^3\delta - yy^3\} \\ & + Q\{\Sigma y\alpha\beta\delta + 3z\alpha^2 y^3\delta - 3w\alpha^2 z^2\delta^2 + 3w\alpha\beta^2 z^2\delta - z^2\delta^2 y^2 + 3xy^3\alpha\gamma\} \\ & + R\{xw - z^2\alpha^3\delta + 4(\alpha^3\delta - \alpha\beta^3) + 3w\alpha\beta y^3 - z\alpha\beta^3 y - y^3\} \\ & + \Theta\{w^2\alpha^2\beta^2 + 2\alpha^2\beta^2\gamma + 2xw + \alpha\beta\delta \cdot 22\alpha\beta\delta + (4z + 8yz)\alpha\beta\gamma\delta + \dots\},\end{aligned}$$

which may be written as follows:

$$\begin{aligned}\Omega = & P\{y(\alpha\delta^3 - \beta^3) + x(\gamma^3 - \beta\delta^3)\} + P(3w\alpha y^3) \\ & + Q\{3z(\beta^3 - \alpha\gamma^2 y^3) + 3w(\alpha^2\delta - \alpha\gamma^2 y) - 3y\delta(\alpha^2\beta\delta - \alpha\beta^2\gamma)\} + Q(3z\alpha\gamma^2 y - 3z\alpha^2\beta^2 y) \\ & + R\{xw + 3z(\alpha^3\delta - \alpha\beta^3) + (\alpha^3\gamma - \alpha\beta^3)\} + R(-3z\alpha^3\gamma) \\ & + \Theta[z w(-\alpha^2\beta\delta + \alpha\beta^2\gamma) \\ & \quad + xz(-\alpha\beta^2\delta + \beta^2\gamma) \\ & \quad + yw(-2\alpha\beta^2\delta + \alpha\beta^2\gamma) \\ & \quad + xw 2(-\alpha\beta\gamma\delta + \alpha\beta\gamma^2) \\ & \quad + yz(-2\alpha^3\delta + 5\alpha^2\beta\delta - 6\alpha\beta^2\gamma) \\ & \quad + x^2(\alpha\beta^3\delta - \alpha\beta^2\gamma^2) \\ & \quad + y^2(-\alpha^4\delta + \beta^4) \\ & \quad + z^2(-\alpha^3\delta + \beta^3\delta) \\ & \quad + w^2(\alpha^2\beta^2 - \beta^3)] ,\end{aligned}$$

in which all the terms contained in the $\{ \}$ admit of expression in terms of P', Q', R', Θ' , the remaining six terms not included within $\{ \}$ may be written

$$\begin{aligned}3wP\alpha(\gamma' - \beta\delta') + 3(wP - yQ)\alpha\beta\Theta' - 3w\alpha\gamma^2\beta\delta'; \\ 3zR\beta(\beta' - \alpha'\gamma) + 3(-xR + zQ)\alpha\gamma^2 - 3\theta wxyz\gamma^2;\end{aligned}$$

which, observing that $wP - yQ = xz\Theta'$, and $-xR + zQ = yw\Theta$, are

$$\begin{aligned}3wP\alpha\beta(\gamma' - \beta\delta') + 3xz\Theta\alpha\beta(\gamma\beta - \beta\delta) + 3(\beta^3 - \alpha\gamma^3) \\ - 3zR\beta(\beta' - \alpha'\gamma) \\ - (\alpha^2\gamma\delta - \alpha\beta\gamma^2)\Theta.\end{aligned}$$